

Training Configuration and Summary:

Dataset used: Training: rssi_odom_dataset_11apr25_isolado.parquet | Experiment: rssi_odom_dataset_12apr25_isolado.parquet

Params:

- model_type: GRU
- group_sec: 100
- timesteps_orig: 60
- batch_size: 1024
- min_units: 64
- max_units: 256
- units_step: 64
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh', 'relu', 'elu']
- optimizers: ['adam', 'rmsprop', 'Nadam']
- min_layers: 1
- max_layers: 2
- max_epochs: 20
- hyperband_iterations: 3
- group_msec: 100
- samples_per_sec: 10
- forecast_horizon_sec: 3.0
- forecast_steps: 30
- timesteps: 60

Tuning:

Trials executed: 10/10

Training Performance:

Loss: 0.0052 | MAE: 0.0558

Validation Performance:

Loss: 0.0086 | MAE: 0.0733

Direct (multi-output) Performance:

MAE: 2.2358 | MSE: 8.1647 | RMSE: 2.8574 | R2: 0.7593

Multi-Step Performance (aggregate):

Steps: 30 (~3.0 s) | MAE: 0.0539 | MSE: 0.0046 | RMSE: 0.0677 | R2: 0.9022

Multi-Step (denormalized, all steps combined):

MAE: 1.95 | MSE: 6.20 | RMSE: 2.49 | R2: 0.759

Per-step (denormalized):

Step 1 → MAE: 1.26 | MSE: 2.50 | RMSE: 1.58 | R2: 0.903

Step 30 → MAE: 2.43 | MSE: 9.42 | RMSE: 3.07 | R2: 0.636

Best Hyperparameters:

Layers: 1 | Activation: elu | Optimizer: adam | Dropout: 0.2

Timing:

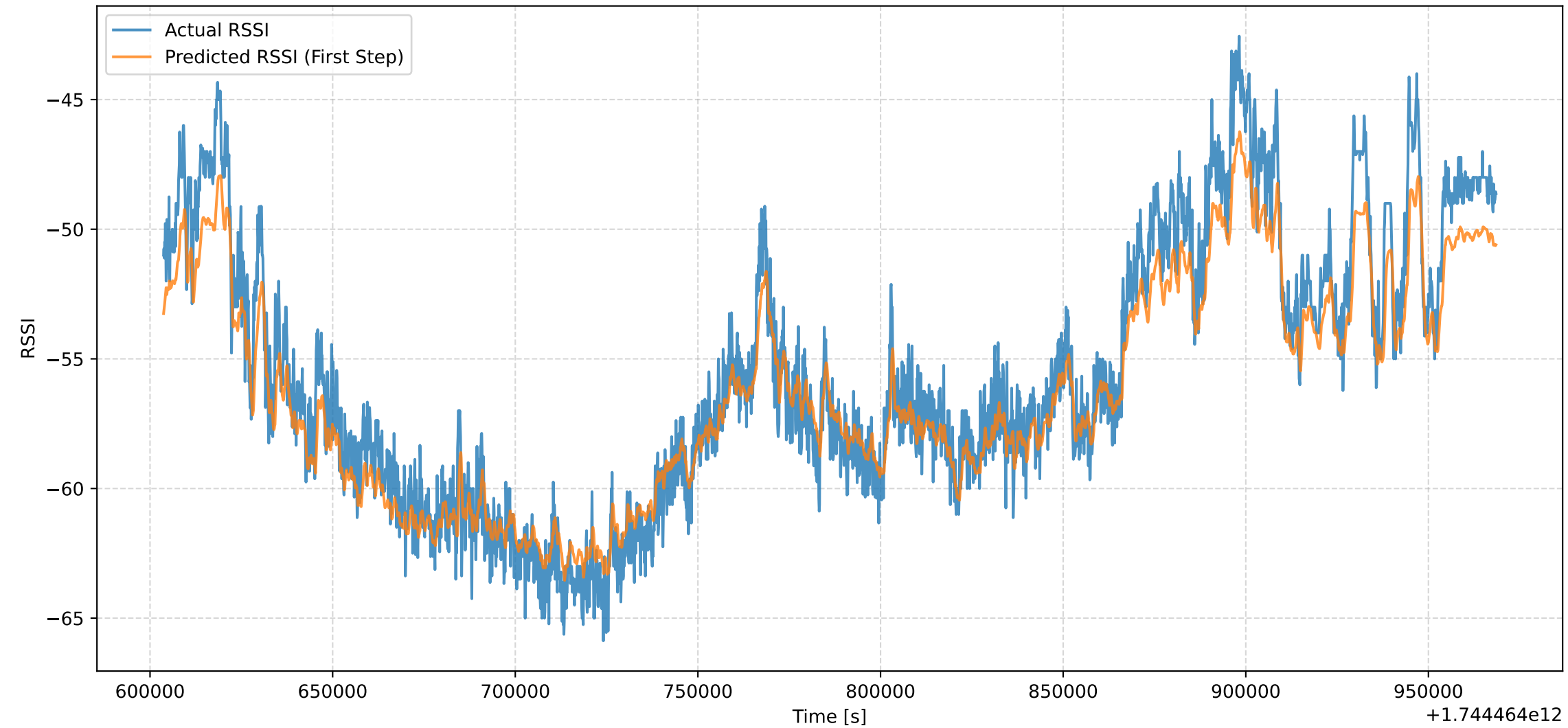
Train: 2025-09-25_06-02-19 → 2025-09-25_06-06-21 (4.02 min)

Predict: 2025-09-25_06-06-21 → 2025-09-25_06-08-29 (2.14 min)

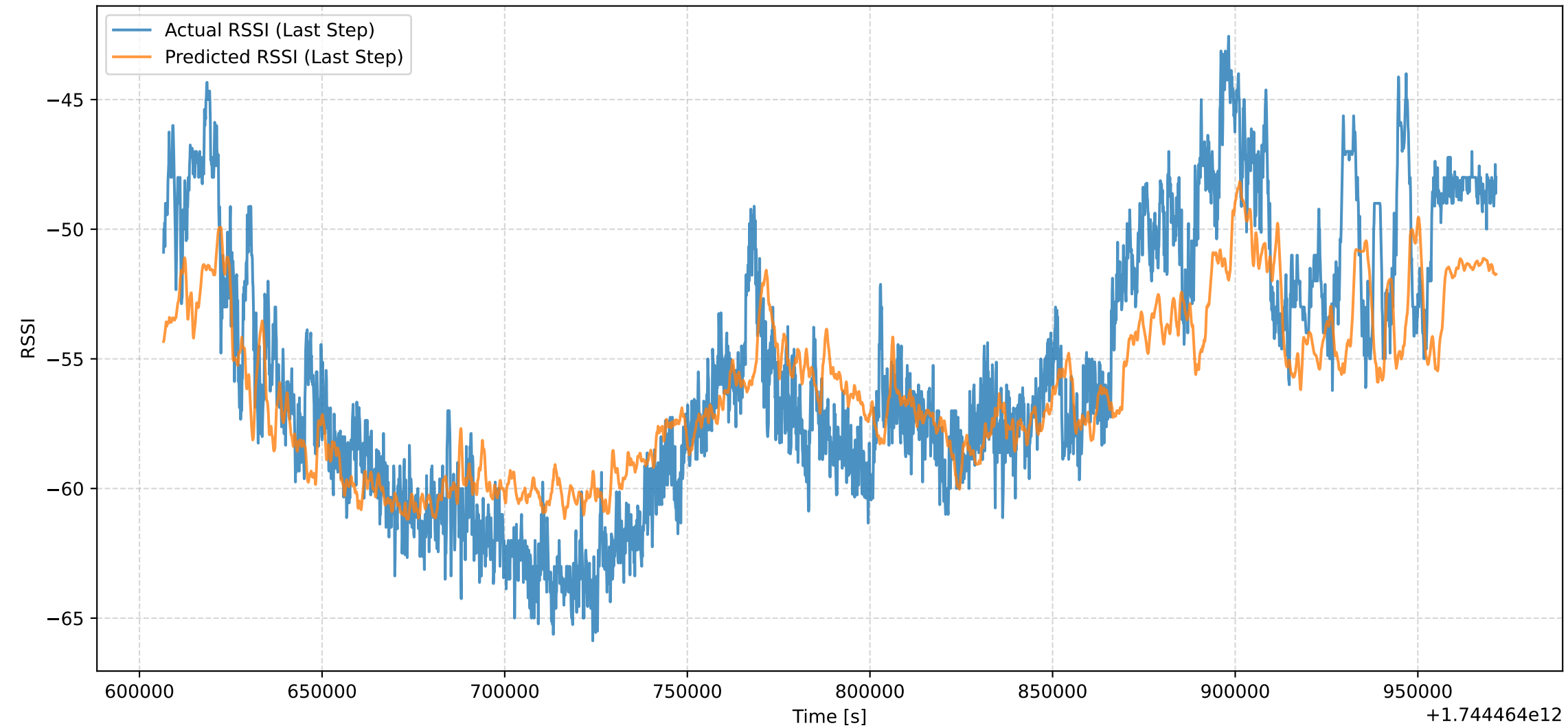
Training dataset down-sample:
Rows before: 40318 | after: 4847

Experiment dataset down-sample:
Rows before: 31097 | after: 3738

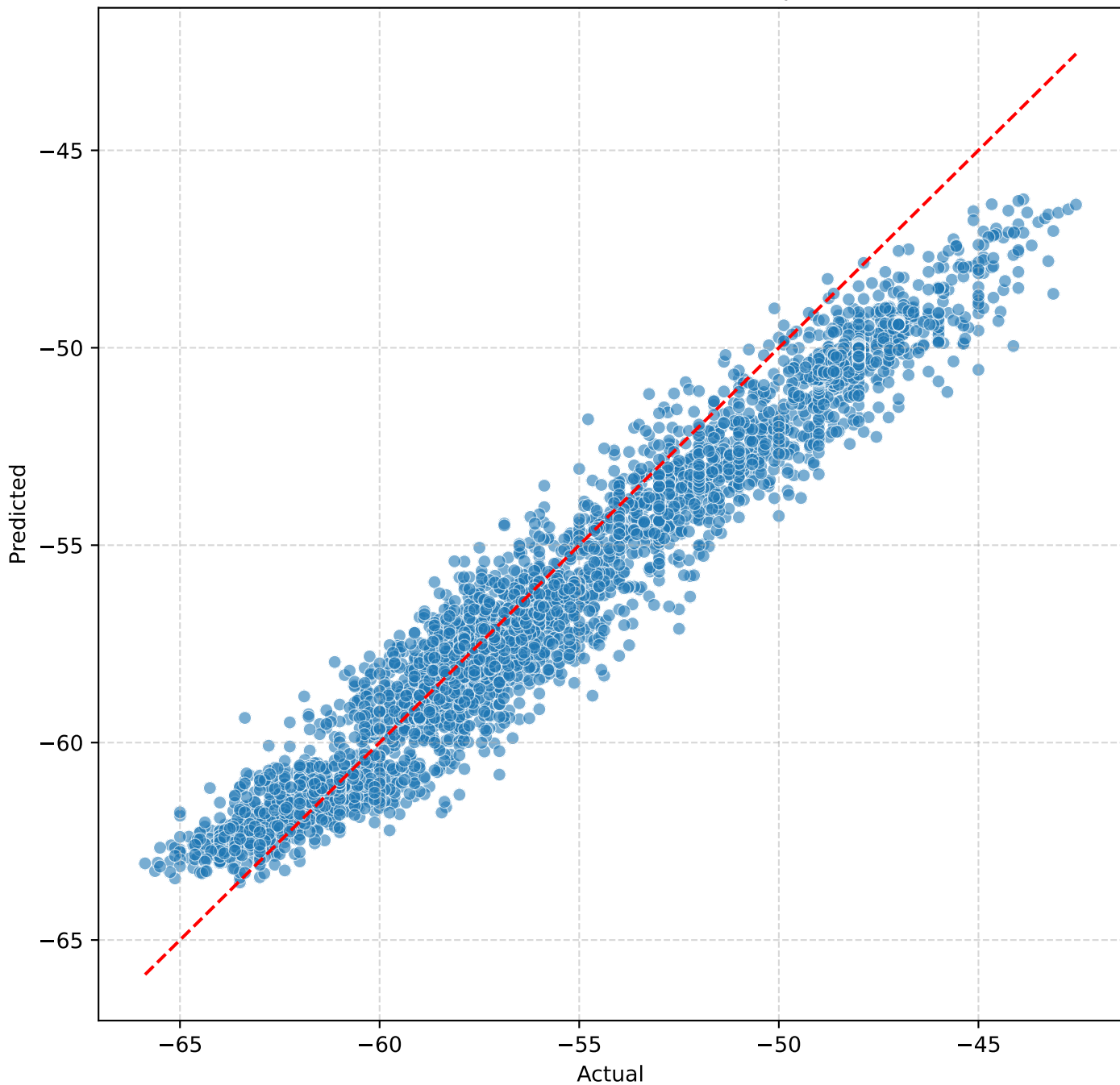
Actual vs Predicted RSSI (First Step)



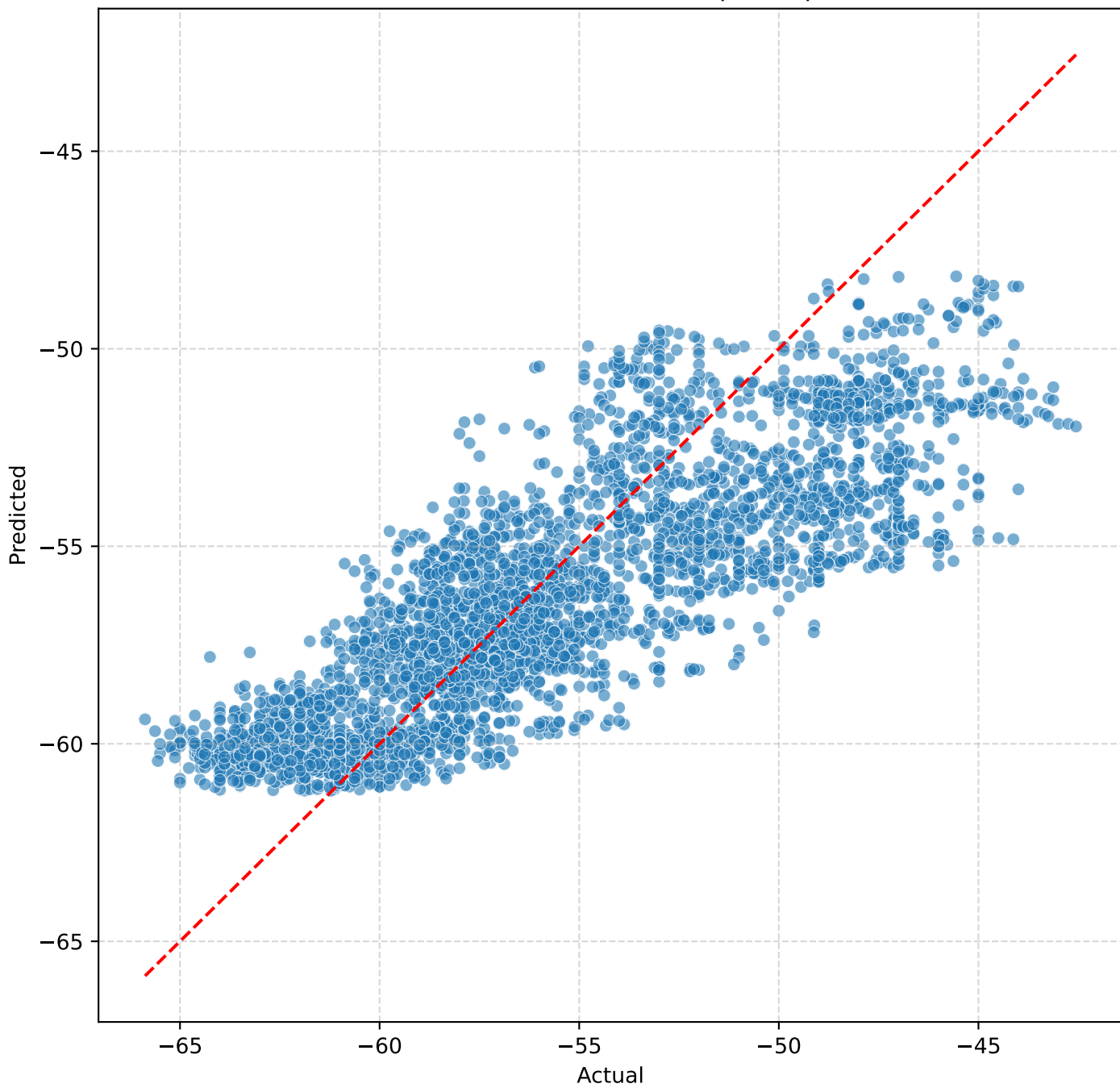
Actual vs Predicted RSSI (Last Step - step 30 $\approx$ 3.0 s)



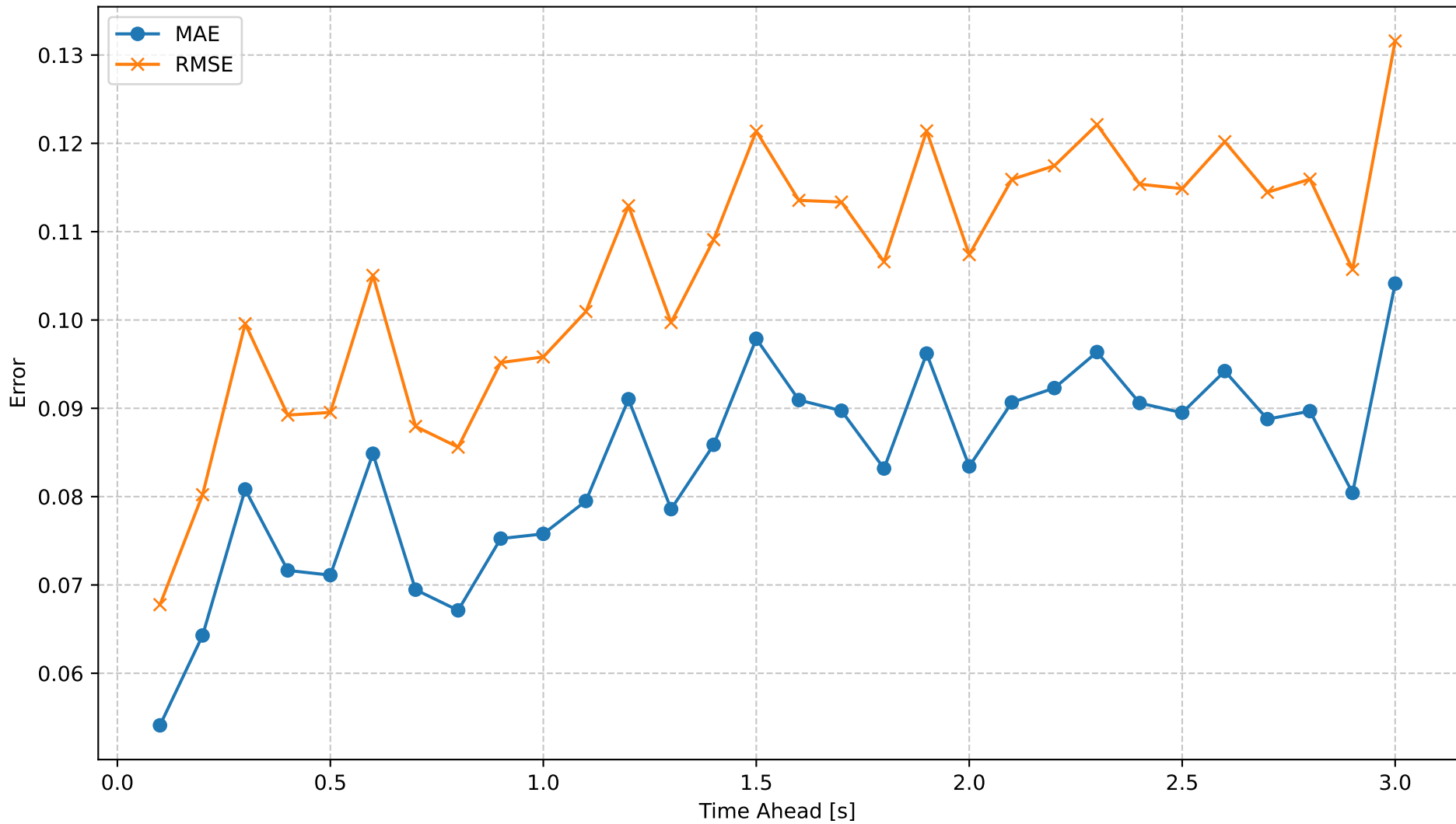
Actual vs Predicted (First Step)



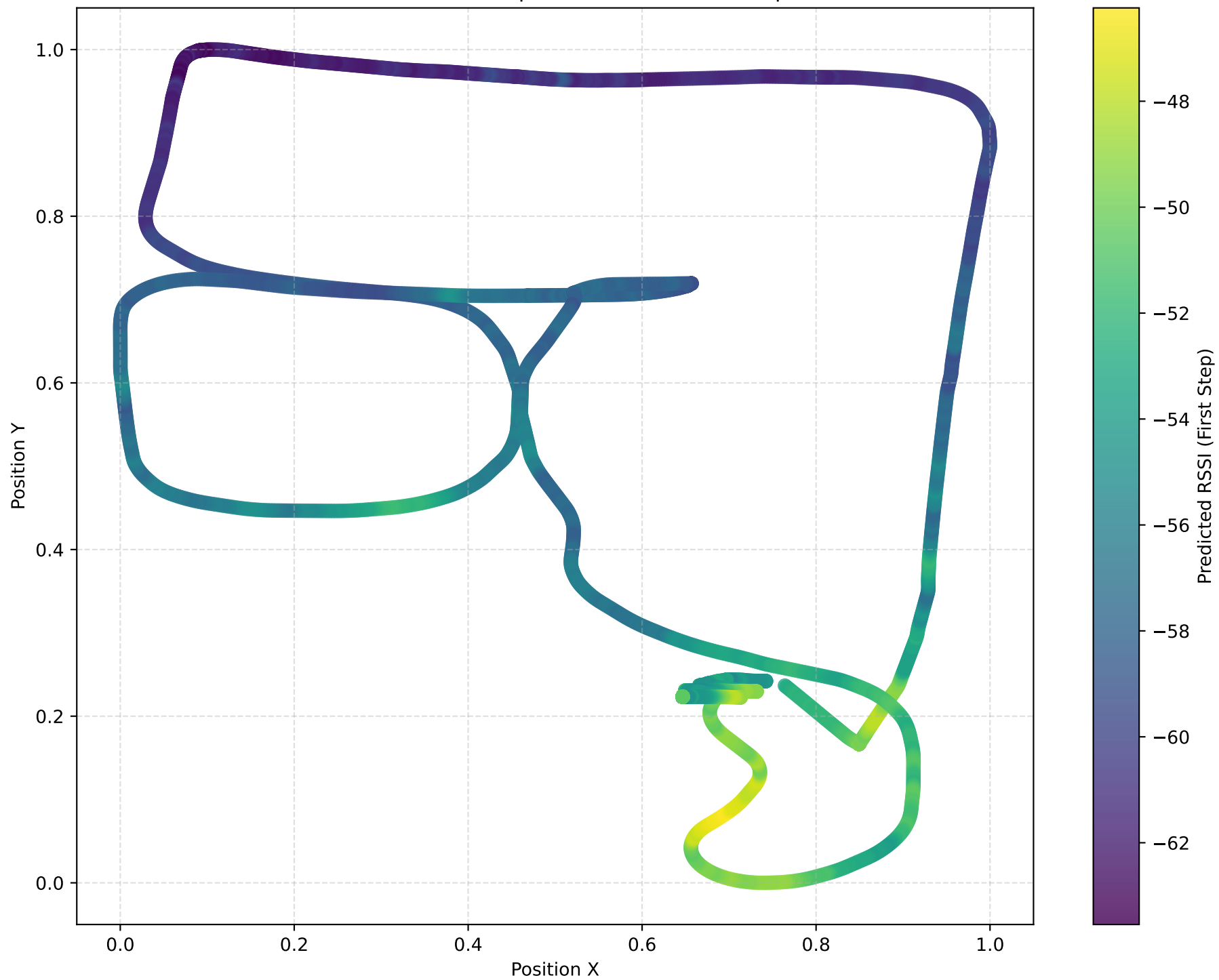
Actual vs Predicted (Last Step - step 30)



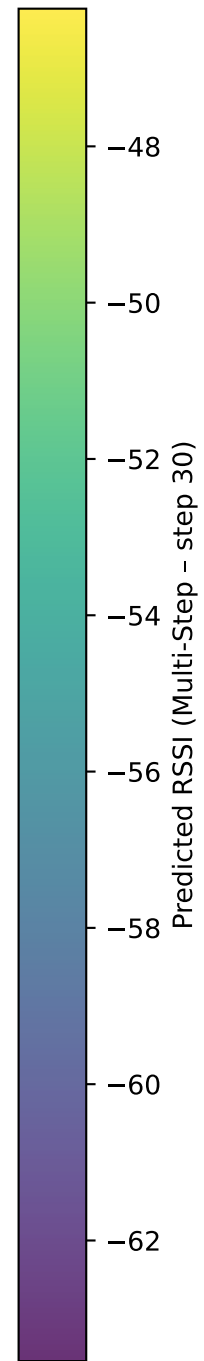
Multi-Step Forecast Error by Time Ahead



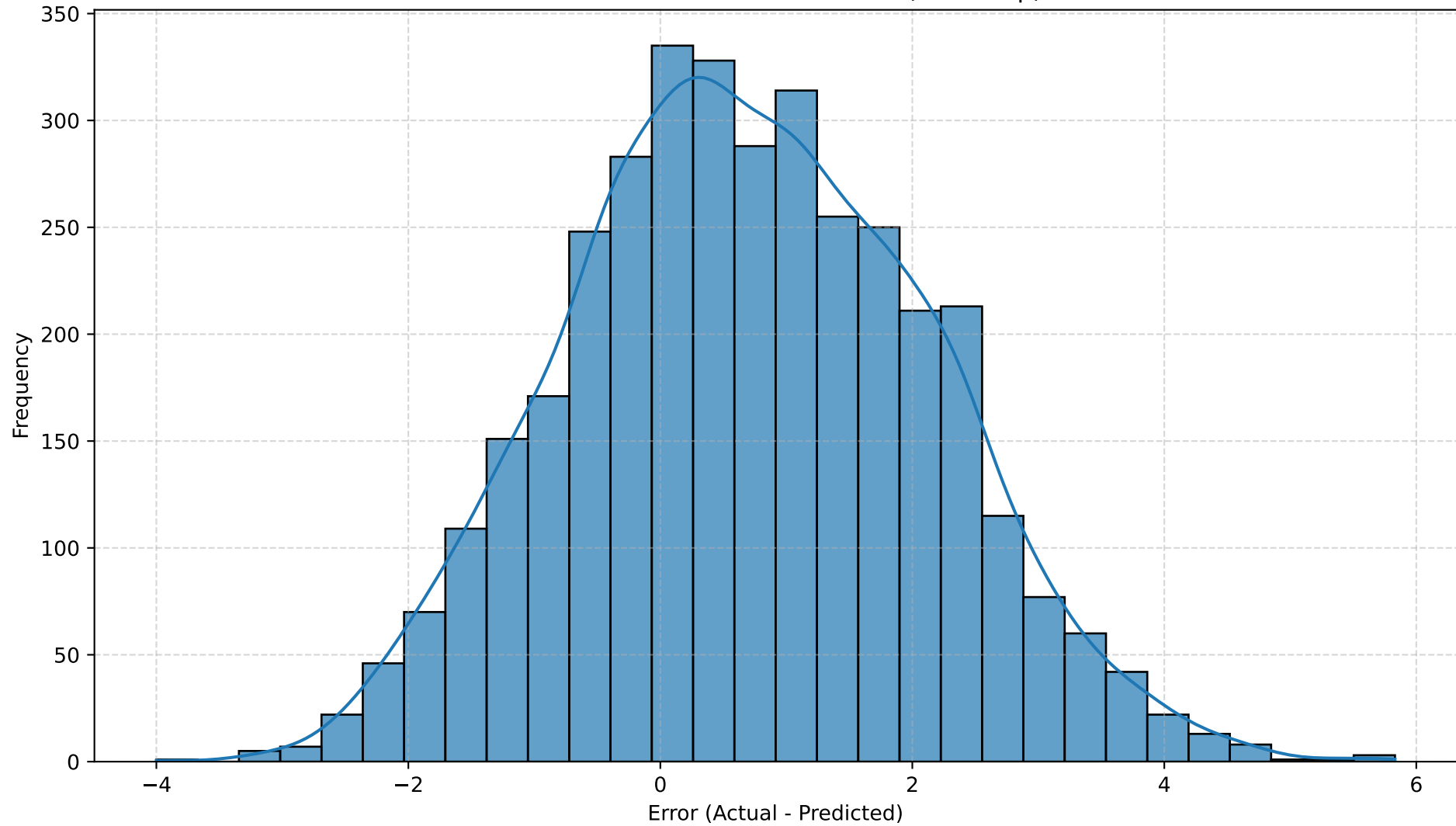
Predicted RSSI in Spatial Context (First Step)



A plot of a complex, self-intersecting curve on a grid. The curve is colored with a gradient from dark blue to light green. It starts at the top left, loops around, crosses itself, and ends at the bottom right. The grid has major lines at 0.0, 0.2, 0.4, 0.6, 0.8, and 1.0 on both axes. The curve is composed of several segments that intersect each other, forming a complex shape. The color gradient suggests a parameterization of the curve, with darker blue representing one end and lighter green representing the other.



Distribution of Prediction Errors (First Step)



Distribution of Prediction Errors (Last Step - step 30)

